

Stat 437 In-Class Mini-Project: Meta-Regression and Blood Pressure

Collins et al. (1990): blood pressure reduction and stroke

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In-Class Mini-Project

This project uses the published overview in Collins et al. (1990) to study a simple meta-regression question:

In medicines that control blood pressure, do trials with larger reductions in diastolic blood pressure (DBP) show larger reductions in stroke risk?

We will use trial-level information from the tables in the paper. The main goal is to practice:

- constructing study-level effect sizes,
- fitting a random-effects meta-analysis,
- fitting a meta-regression,
- and interpreting the moderator(s) in context.

A small warning before we start: the study-level moderator is not the same thing as a patient-level causal effect. Keep that in mind as you work.

Data entry

The table below combines the trial characteristics from Tables 1 and 2 with the stroke outcome counts from the overview figure on page 831. Carter did not report the DBP difference in attendees, so that value is left as missing.

Variable Descriptions

The dataset (names `collins_bp` here) combines trial characteristics and outcomes from Collins et al. (1990). Each row corresponds to a randomized trial (or subgroup within a trial).

- **trial**: Name of the study or trial identifier.
- **trial_group**: Description of the study population or subgroup (often based on baseline diastolic blood pressure categories or trial design).
- **age**: Mean age of participants in the study (years).
- **male_pct**: Percentage of participants who are male.
- **follow_up**: Average duration of follow-up (years).
- **blinding**: Blinding design of the trial (e.g., double-blind, single-blind, or none).
- **entry_dbp**: Range or threshold of diastolic blood pressure (DBP) used as an inclusion criterion at baseline (mm Hg).
- **dbp_diff**: Difference in diastolic blood pressure between treatment and control groups during the trial (mm Hg). *This is the key moderator used in the meta-regression.*
- **e_t**: Number of stroke events observed in the treatment group.

- **n_t**: Total number of participants in the treatment group.
- **e_c**: Number of stroke events observed in the control group.
- **n_c**: Total number of participants in the control group.
- **main_drugs**: Primary antihypertensive medications used in the treatment group (abbreviations from the original paper, e.g., diuretics, beta-blockers, etc.).

Trial	Age	Male %	Follow (yr)	Blinding	Entry DBP	DBP diff	Stroke Tx	Tx N	Stroke Ctl	Ctl N	Drugs
VA-NHLBI	38	81	1.5	Double	85-105	7	0	508	0	504	CD
HDFP	51	55	5.0	None	mixed	5	102	5485	158	5455	CD
Oslo	45	100	5.5	None	90-109	10	0	406	5	379	HZ
ANBPS	50	63	4.0	Single	95-109	6	13	1721	22	1706	CZ
MRC	52	52	5.0	Single	95-109	6	60	8700	109	8654	BF or PR
VA (90-114)	51	100	3.3	Double	90-114	19	5	186	20	194	HZ + RE + HD
USPHS	44	80	7.0	Double	90-114	10	1	193	6	196	CZ + RU
HSCSG	59	41	2.3	Double	90-115	12	43	233	52	219	MC + DS
VA (115-129)	51	100	1.5	Double	115-129	27	5	70	20	73	HZ + RE + HD
Wolff	49	32	1.4	Double	100-130	20	2	45	1	42	RE
Barracough	56	43	2.0	Single	100-120	13	0	58	0	58	BF or MD
EWPHS	72	30	4.7	Double	90-119	10	32	416	48	424	HZ + TT
Coope	69	31	4.4	None	105-120	11	20	419	39	465	AT

Part 1. Build study-level effect sizes

For the rest of the project, use the stroke counts to compute a study-level log odds ratio and its sampling variance.

Use `metafor::escalc()` with `measure = "OR"`.

Something to think about:

- VA-NHLBI has 0 events in both arms.
- What does a continuity correction do in a double-zero trial?
- Are we comfortable adding a number that isn't connected to data?
 - Should this study stay in the primary analysis, or should it be handled as a sensitivity analysis?

Answer Here

```
# Create the 2 x 2 table columns needed by escalc()
# ai = events in treatment
# bi = non-events in treatment
# ci = events in control
# di = non-events in control

dat_es <- escalc(
  measure = "OR",
  ai = e_t,
  bi = n_t - e_t,
  ci = e_c,
  di = n_c - e_c,
  data = collins_bp,
  add = 0.5,      ## this is a continuity correction for the 2x2 table
  ## this continuity correction is only applied to all.
  to = "all",
  drop00 = FALSE
)
```

```
# Inspect the study-level log(OR) values and variances
dat_es %>%
  dplyr::select(trial, dbp_diff, yi, vi)
```

	trial	dbp_diff	yi	vi
1	VA-NHLBI	7	-0.0079	4.0039
2	HDFP	5	-0.4520	0.0164
3	Oslo	10	-2.4799	2.1869
4	ANBPS	6	-0.5250	0.1197
5	MRC	6	-0.6043	0.0259
6	VA (90-114)	19	-1.3550	0.2418
7	USPHS	10	-1.4768	0.8310
8	HSCSG	12	-0.3167	0.0533
9	VA (115-129)	27	-1.5180	0.2646
10	Wolff	20	0.4638	1.1138
11	Barracclough	13	0.0000	4.0342
12	EWPHS	10	-0.4213	0.0566
13	Coope	11	-0.5905	0.0789

Questions:

- What is the interpretation of yi?

yi is the “effect size” that the study found, expressed as a log odds ratio between the odds of getting a stroke in the treatment group over the control

- What does vi represent?

vi is the variance of that yi, which we’re going to use to weight our studie

- Which studies look most extreme?

The Olso, the VA-NHLBd, and the Barracclough studies all have the most extreme effect sizes.

- Which study has the largest sample size, and why might that matter?

The MRC study was the biggest, which matters because it means that it’ll have the lowest variance and the most weight later on, defining a large part of our meta regression.

Part 2. Fit a fixed-effects model

Fit a fixed-effects model to establish the baseline heterogeneity.

```
re_mod <- rma(yi = yi, vi = vi, data = dat_es, method = "FE")
summary(re_mod)
```

Fixed-Effects Model (k = 13)

logLik	deviance	AIC	BIC	AICc
-9.2183	12.0334	20.4366	21.0015	20.8002

I² (total heterogeneity / total variability): 0.28%

H² (total variability / sampling variability): 1.00

Test for Heterogeneity:

Q(df = 12) = 12.0334, p-val = 0.4430

Model Results:

estimate	se	zval	pval	ci.lb	ci.ub	
-0.5357	0.0772	-6.9403	<.0001	-0.6869	-0.3844	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Questions:

- What is the estimated average effect?

The average log odds was -0.5357

- What are the estimated τ^2 and I^2 values?

The I^2 estimate is 0.28%, which is basically zero, and tau squared is basically zero as well, according to people in class.

- Is heterogeneity negligible, moderate, or substantial?

The heterogeneity is basically negligible.

- Do you think a random-effects model is needed here? Why or why not?

This doesn't make me think that we actually need the random-effects model because it seems like all of the studies are measuring about the same thing, since the variability due to heterogeneity is low.

Part 3. Fit a meta-regression with DBP reduction

Now ask whether the amount of DBP reduction helps explain the variation in stroke risk across trials.

```
meta_mod <- rma(
  yi = yi,
  vi = vi,
  mods = ~ dbp_diff,
  data = dat_es,
  method = "FE"
)

summary(meta_mod)
```

Fixed-Effects with Moderators Model (k = 13)

logLik	deviance	AIC	BIC	AICc
-7.8983	9.3934	19.7966	20.9265	20.9966

I^2 (residual heterogeneity / unaccounted variability): 0.00%

H^2 (unaccounted variability / sampling variability): 0.85

R^2 (amount of heterogeneity accounted for): 14.84%

Test for Residual Heterogeneity:

QE(df = 11) = 9.3934, p-val = 0.5856

Test of Moderators (coefficient 2):

QM(df = 1) = 2.6400, p-val = 0.1042

Model Results:

	estimate	se	zval	pval	ci.lb	ci.ub
intrcpt	-0.3061	0.1610	-1.9015	0.0572	-0.6216	0.0094
dbp_diff	-0.0286	0.0176	-1.6248	0.1042	-0.0631	0.0059

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Questions:

- What is the sign of the slope for `dbp_diff`?

The slope is negative.

- Does the direction make scientific sense?

This makes scientific sense, because higher values of diastolic blood pressure (better blood pressure) mean a lower log odds ratio with means that there is lower chance of stroke in the treatment group than the control group.

- How should you interpret the moderator in context?

I would interpret it as the way the treatment (the drugs) affect the outcomes (the numbers of strokes).

- What does the test for moderators tell you?

It tells us if the moderator is actually acting as a significant connection between the treatment and the control.

Now compare the heterogeneity before and after adding the moderator.

```
tibble(
  model = c("Random effects", "Meta-regression"),
  tau2 = c(re_mod$tau2, meta_mod$tau2),
  I2 = c(re_mod$I2, meta_mod$I2),
  R2_analog = c(NA, meta_mod$R2)
)
```

A tibble: 2 x 4

	model	tau2	I2	R2_analog
	<chr>	<dbl>	<dbl>	<dbl>
1	Random effects	0	0.278	NA
2	Meta-regression	0	0	14.8

Questions:

- How much of the heterogeneity is explained by `dbp_diff`?

0.2775625

- What does the R2 analog mean here?

It is the percentage of the variation in the log odds between the treatment and the control that is explained by the meta regression model.

- Is this the same as ordinary regression R^2 ?

Not exactly, because this is tracking the variation between the studies that is explained, not the variation between the individuals.

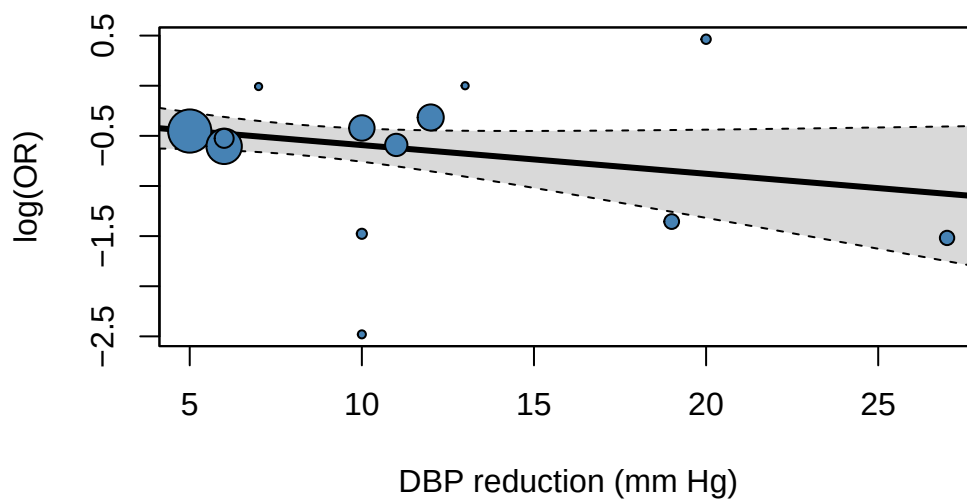
- If R2 is small, does that mean the moderator is unimportant?

I would say that it is still important because a difference in 14% in the response could be the difference between life and death.

Part 4. Visualize the meta-regression

Make a bubble plot of the fitted meta-regression.

```
regplot(  
  meta_mod,  
  xlab = "DBP reduction (mm Hg)",  
  ylab = "log(OR)",  
  pch = 21,  
  bg = "steelblue"  
)
```



Questions:

- Which studies have the most leverage in the plot?

It looks like the outliers are having the most leverage on the plot.

- Why do a few small studies with extreme effects sometimes influence the fitted line so much?

They influence the line a lot because even though they carry low weights, because they're so far off they're exerting a lot of influence on the slope of the line.

- Does a strong line mean the relationship is causal?

Not necessarily, a really good regression still is only showing correlation between variables on a study by study basis, correlations that could be driven by other underlying factors.

Part 5. Discussion

Discuss these questions with your group:

1. The ecological fallacy occurs when relationships observed between variables at an aggregate (group) level are incorrectly assumed to hold true for individuals, leading to biased conclusions. It is a major risk in meta-regressions using aggregated data. Here, if studies with larger DBP reductions also have lower stroke risk, can we conclude that individual patients who lower DBP more will necessarily benefit more? Why or why not?

We can't make that conclusion because we're still subject to Simpson's paradox or the ecological fallacy here. Individual studies could show the opposite from what the regression of studies shows.

2. Leverage: which studies are most influential in this meta-regression, and why might small trials with large effects be especially influential?

The small influential studies on the fringes of the visualization are the most influential, which will often be small trials with large effects.

3. Interpretation of R^2 : If R^2 is 0% but the moderator is statistically significant, what does that tell us about the 'unexplained' variance?"

Dr. White said that we didn't have to answer this one.

4. Model justification: if `dbp_diff` explains little heterogeneity, what other study-level moderators might you try next, and which ones would you trust most?

Age could be a good one to try, since there's some real variability there, as well as the average duration to follow up which can capture greater effects over time.

Optional extension if time remains:

- Try another moderator/variable such as `age` or `male_pct`.
- Does it explain more or less heterogeneity than `dbp_diff`?
- Would you trust it as a more plausible scientific explanation? ““